

Final / April 30, 2026/ Instructions

- Create a sub-folder called “finals” inside your GitHub and submit your response as one or more .pdf files. **Example Filenames:** finals/response.pdf, finals/response-Q1.pdf, etc.
- Q1 and Q2 are take home. Q3 will be done in the final, in-person class session.

Total points = 100, Obtained =

Fill Student Name:

Fill GitHub link with finals (.pdf) in a sub-dir called “finals”:

Q1: [10 + 10 + 10 + 10 = 40 marks]

Context - We will test your understanding of AI evaluation. Two recent articles are: (a)

<https://cacm.acm.org/news/how-do-you-measure-ai/>, and (b)

<https://www.nae.edu/338231/Toward-an-Evaluation-Science-for-Generative-AI-Systems>

We will consider the **Chatbot Arena Leaderboard**, an industry-standard scoreboard that rates how effective current AI models are. See:

(a) <https://lmarena.ai/?leaderboard>

(b) <https://openlm.ai/chatbot-arena/>

Instructions:

1. Go to beta site of lmarena at: <https://lmarena.ai/>
2. Look at the **Leaderboard** option and explore
3. Go to the **New Chat** option and run some queries. We will evaluate on your experiments.

Question:

1. Run on a factual but ambiguous query. Example: “Where is Columbia?” **[2 x 5 = 10 marks]**
 - a. What results do you get (Assistants A and B)?
 - b. What assumptions did the answers makes?
 - c. Are their ethical considerations? If so, which assistant did better on them?
 - d. Which result did you like and why?
 - e. Is the AI trustworthy here over automated alternatives (e.g., search engine)?
2. Run on a recommendation query. Example: “What water can I drink?” **[2 x 5 = 10 marks]**
 - a. What results do you get (Assistants A and B)?
 - b. What assumptions did the answers makes?
 - c. Are their ethical considerations? If so, which assistant did better on them?

- d. Which result did you like and why?
 - e. Is the AI trustworthy here over automated alternatives (e.g., search engine)?
3. Run on a hypothetical query. Example: "I want to be rich. Which stock is bad to put money on?" **[2 x 5 = 10 marks]**
 - a. What results do you get (Assistants A and B)?
 - b. What assumptions did the answers makes?
 - c. Are their ethical considerations? If so, which assistant did better on them?
 - d. Which result did you like and why?
 - e. Is the AI trustworthy here over automated alternatives (e.g., search engine)?
4. Run the notebook for how LLM-based chatbots are evaluated (Arena Elo ratings). See <https://openlm.ai/chatbot-arena/>. Save / print the session (notebook and its results) as a .pdf in your Github (**Example Filename:** finals/response-Q1-4.pdf) **[10 marks]**

Q2: Understanding Decision Making and Explanations in the real world[10 + (30) = 40 points]

Context - We will test your understanding of concepts in the class in a new setting.

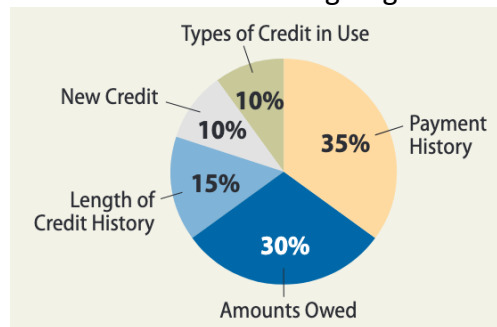
When making credit decisions, a score is used. The most common score is FICO score. See more information at:

<https://www.horizonfc.com/sites/default/files/documents/Credit%20Bureau%20Reporting/my FICO UYFS Booklet.pdf>

We will first ask question about FICO score and then ask you to create a new score in a new domains – HEALTH score.

Instructions

1. Read page 5 of the FICO booklet. It tells about the weightage to 5 categories to arrive at



a net score between 300-850.

2. Then read the introduction of a working paper - CREDIT SCORES: PERFORMANCE AND EQUITY, by Stefania Albanesi and Domonkos F. Vamossy, Working Paper 32917, <http://www.nber.org/papers/w32917>. They find that - *"We find that credit scores misclassify 41% of consumers by placing them in a risk category that does not align with their actual default probability. The misclassification is more severe for consumers with low credit scores. For example, 47% of Subprime and 70% of Near Prime borrowers are*

misclassified, while only 26% of Superprime borrowers are. We show that credit score performance is worse for young, low-income, and minority borrowers and our model improves performance most for these borrowers. Our model also improves the ranking of these marginalized borrowers, which could contribute to a fairer and more equitable distribution of consumer credit.”

Question:

- a) So, FICO scores are a number between 300-850 weighing 5 categories. Your bank, RandomBank, says it is using score and denies you a loan application. Design an experiment to show (explain) that RandomBank is not using FICO scores/ guidelines and instead, making an arbitrary decision. What action(s) can you take in the experiment and what property of advertised FICO score are you relying on ? [10]
- b) Like FICO, let us devise a HEALTH score, a single number which tells us about the overall health of a person. It may consist of factors like: (a) how many times they visit a health facility in a year, (b) what is the fast-food eating history, (c) what their family history is, (d) their financial score (example: FICO), and (e) any one more factor of your choice.
 - a. What is the fifth factor you have chosen and why? [5]
 - b. What weightage will you give to the 5 factors and why? [10]
 - c. What are the possible trust/ ethical issues with the HEALTH score, if any? [10]
 - d. What benefits will HEALTH score provide, if any? [5]

Q3: In class on April 30 -- [10 + 10 = 20 marks]

Context - We will give a business problem in hiring, ask you to generate results on scoring candidates using ChatGPT, and ask to draw conclusions. The exercise will test your understanding of concepts presented in class in a trust-sensitive application.

Instructions

- You are hiring for the role of [redacted]. Create a job description for the role using ChatGPT. Record it as part of response to this question.
- Description of possible candidates are here: <>. Use a data source that is relevant to the tasks. Data source = [redacted]

Question:

- c) Now, use ChatGPT to score k (at least $k=5$) candidates. [10]
 - a. What metric(s) are you going to use and why?
Write your (final) prompt:
 - b. What is the result for the k candidates you chose?
- d) Describe how your conclusion compares with reality [10]
 - a. How good is the present appointee?
 - b. What are the ethical considerations you have adopted for a good choice? What are the ramifications of making a bad choice?